

```

async function predict() {
  // Prediction #1: run input through posenet
  // estimatePose can take in an image, video or canvas html element
  const { pose, posenetOutput } = await model.estimatePose(webcam.canvas);
  // Prediction 2: run input through teachable machine classification model
  const prediction = await model.predict(posenetOutput);

  for (let i = 0; i < maxPredictions; i++) {
    const classPrediction =
      prediction[i].className + " : " + prediction[i].probability.toFixed(2);
    labelContainer.childNodes[i].innerHTML = classPrediction;
    if (prediction[3].probability.toFixed(2) >= 0.98 || (prediction[4].probability.toFixed(2) >= 0.98 || (prediction[5].probability.toFixed(2) >= 0.98)) {
      document.getElementById("output").innerHTML = "Suspicious activity";
      var audio = new Audio("/home/pi/siren.mp4");
      audio.play();
    } else {
      document.getElementById("output").innerHTML = "Normal activity";
    }
    // document.getElementById("output").innerHTML = prediction[2].className + prediction[2].probability.toFixed(2);
  }
}

```

Fig. 8: Code for play alert sound

Ashwini Sinha AI suspicious activity detector camera

Recognise Movement
Suspicious Activity



Fig. 9: Testing and output of camera

Any of these can be chosen for the project. To demonstrate, Google Teachable is being used here.

In Google Teachable, select the PoseNet option for tracking. Correctly various body movements and actions. Correctly label the actions performed earlier of walking, talking, eating, standing, etc. Feed these datasets into the ML model. Similarly, correctly label and feed the datasets of activities like pointing guns, firing guns, fighting, beating, etc.

After feeding the required datasets, click on Train Model. The option to export the ML model or upload it to the cloud will appear. Uploaded the ML model to the cloud to safely store it there and let users access and deploy it anywhere in the world.

On uploading to the cloud, you will receive a URL for using the ML

model. The URL of the ML model created by the author is: <https://teachablemachine.withgoogle.com/models/jLeJT1rQ/>

Deploy the ML model on the camera and connect a speaker so that the Raspberry Pi triggers an

alarm when any suspicious activity gets detected.

Coding

To deploy the ML model in Raspberry Pi coding is required. While uploading the ML model you would have received an example code snippet. Copy and paste it into your JavaScript (JS) code. JS coding and testing can be done with the help of any code editor, offline or online. You can name the camera in HTML for the ML model.

Import the ML model that was previously uploaded to the cloud. Then check whether activities such as beating, gun firing, and gun possession are detected by the ML model during video processing.

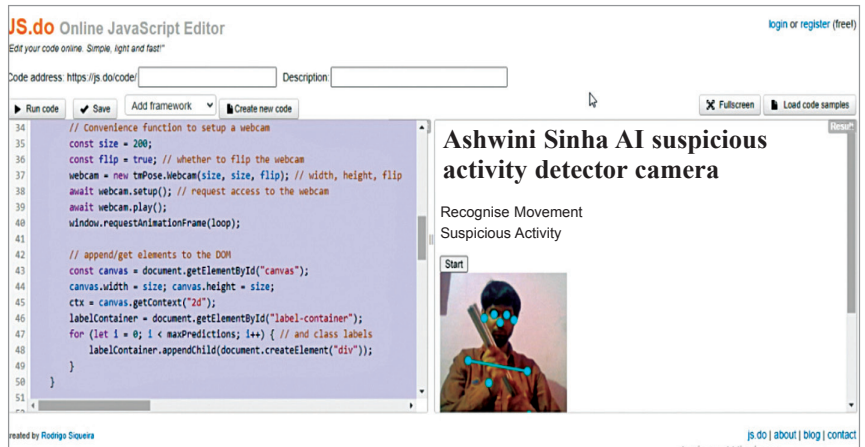


Fig. 10: Testing in .js online editor and compiler

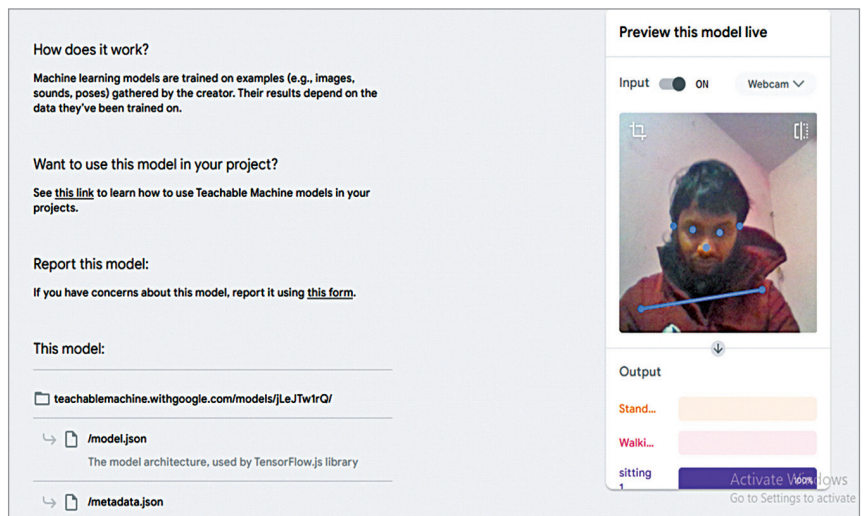


Fig. 11: Testing camera using URL of trained ML model